

■ AI Data Analyst - Executive Analysis Report

■ Executive Data Analysis Report

1. Executive Summary

- The dataset contains **12,575** transactions spanning **2022-01-01** to **2025-01-18**.
- **Total revenue** is **\$1,552,071** with an **average transaction value** of **\$129.65** (median **\$108.50**).
- The three highest-earning product categories (**Butchers**, **Electric household essentials**, **Beverages**) generate **≈ \$608 k** (≈ 39 % of total revenue).
- **Cash** is the leading payment method (≈ 35 % of revenue), closely followed by **Digital Wallet** and **Credit Card**.
- Discounts are applied in **33.55 %** of purchases, but the *Discount Applied* column has **4,199 missing values** (≈ 33 %).
- Data quality is solid (no duplicate transaction IDs, no invalid discount flags), yet **significant missingness** exists in *Item*, *Price Per Unit*, *Quantity*, and *Total Spent*.
- **60 outlier transactions** (identified via the IQR method on *Total Spent*) merit further investigation.

Bottom line: The business is driven by a few key categories and cash payments, while discount usage is substantial. Addressing missing data and reviewing outliers will improve analytical reliability and uncover potential revenue opportunities.

2. Dataset Overview & Data Quality Profile

Aspect Detail
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Rows × Columns 12,575 × 11
Duplicate rows 0
Missing values • Item: 1,213 • Price Per Unit: 609 • Quantity: 604 • Total Spent: 604 • Discount Applied: 4,199
Potential date column Transaction Date (converted to datetime)
Invalid entries None in Discount Applied (all True/False/NaN)
Outliers (Total Spent) 60 transaction IDs flagged

3. Data Cleaning & Validation Summary

- **Deduplication:** `df.drop_duplicates()` – no duplicates were removed.
- **Date conversion:** `Transaction Date` parsed to `datetime64[ns]` (coercing errors).
- **Validation status:** `validated` – all rows retained (`Clean Rows = 12,575`).

4. User Query & Analytical Approach

User Question:

> “Analyze key dataset metrics, profile anomalies, and generate summary insights.”

Analytical Methodology (implemented in Pandas):

1. **Core Metrics** – total transactions, revenue, average/median transaction value, total quantity, average price per unit.
2. **Revenue breakdowns** – by `Category` and by `Payment Method`.
3. **Discount analysis** – count of True/False/NaN, discount usage rate.
4. **Date range** – earliest and latest transaction dates.
5. **Anomaly profiling** – missing value counts, duplicate transaction IDs, invalid discount entries, outlier detection on `Total Spent` using the IQR method.
6. **Summary insights** – top 3 categories & payment methods, discount usage, data quality flags.

All steps are performed on the cleaned `df` and the results are stored in the variable `result`.

5. Python Analysis Code & Executed Data Output

```
# Convert Transaction Date to datetime (in-place) df['Transaction Date'] =
pd.to_datetime(df['Transaction Date'], errors='coerce') # ----- Core Metrics -----
total_transactions = len(df) total_revenue = df['Total Spent'].sum() avg_transaction_value =
df['Total Spent'].mean() median_transaction_value = df['Total Spent'].median()
total_quantity_sold = df['Quantity'].sum() avg_price_per_unit = df['Price Per Unit'].mean() #
Revenue by Category revenue_by_category_df = ( df.groupby('Category', as_index=False)['Total
Spent'] .sum() .rename(columns={'Total Spent': 'Revenue'}) .sort_values('Revenue',
ascending=False) ) # Revenue by Payment Method revenue_by_payment_df = ( df.groupby('Payment
Method', as_index=False)['Total Spent'] .sum() .rename(columns={'Total Spent': 'Revenue'})
.sort_values('Revenue', ascending=False) ) # Discount statistics discount_counts_series =
df['Discount Applied'].value_counts(dropna=False) discount_rate =
discount_counts_series.get(True, 0) / total_transactions date_min, date_max = df['Transaction
Date'].min(), df['Transaction Date'].max() # ----- Anomaly Profiling -----
```

```

missing_counts = df.isna().sum().to_dict() duplicate_txn_count = df['Transaction
ID'].duplicated().sum() # Invalid entries in Discount Applied (neither True, False nor NaN)
invalid_discount_mask = ( df['Discount Applied'].notna() & ~df['Discount Applied'].isin([True,
False]) ) invalid_discount_count = invalid_discount_mask.sum() # Outliers in Total Spent using
the IQR method Q1 = df['Total Spent'].quantile(0.25) Q3 = df['Total Spent'].quantile(0.75) IQR
= Q3 - Q1 lower_bound = Q1 - 1.5 * IQR upper_bound = Q3 + 1.5 * IQR outlier_mask = (df['Total
Spent'] < lower_bound) | (df['Total Spent'] > upper_bound) outlier_transactions =
df.loc[outlier_mask, 'Transaction ID'].tolist() # ----- Summary Insights -----
top_categories = ( revenue_by_category_df.head(3) .set_index('Category')['Revenue'] .to_dict()
) top_payment_methods = ( revenue_by_payment_df.head(3) .set_index('Payment Method')['Revenue']
.to_dict() ) insights = { "date_range": f"{date_min.date()} to {date_max.date()}",
"top_3_categories_by_revenue": top_categories, "top_3_payment_methods_by_revenue":
top_payment_methods, "discount_usage_rate": round(discount_rate, 4), "has_missing_values":
any(v > 0 for v in missing_counts.values()), "duplicate_transactions":
int(duplicate_txn_count), "invalid_discount_entries": int(invalid_discount_count),
"total_outlier_transactions": len(outlier_transactions) } # ----- Assemble Result
----- result = { "metrics": { "total_transactions": total_transactions, "total_revenue":
total_revenue, "average_transaction_value": avg_transaction_value, "median_transaction_value":
median_transaction_value, "total_quantity_sold": total_quantity_sold, "average_price_per_unit":
avg_price_per_unit, "revenue_by_category": revenue_by_category_df.to_dict(orient='records'),
"revenue_by_payment_method": revenue_by_payment_df.to_dict(orient='records'),
"discount_counts": discount_counts_series.to_dict(), "discount_rate": discount_rate,
"transaction_date_range": { "start": date_min.isoformat(), "end": date_max.isoformat() } },
"anomalies": { "missing_values_per_column": missing_counts, "duplicate_transaction_ids":
int(duplicate_txn_count), "invalid_discount_entries": int(invalid_discount_count),
"outlier_transaction_ids": outlier_transactions }, "insights": insights }

```

Executed Output (high-level view)

Metric	Value
Total transactions	12,575
Total revenue	\$1,552,071
Average transaction value	\$129.65
Median transaction value	\$108.50
Total quantity sold	66,276
Average price per unit	\$23.37
Discount usage rate	33.55 %
Date range	2022-01-01 → 2025-01-18
Missing values (any column?)	Yes
Duplicate transaction IDs	0
Invalid discount entries	0
Outlier transactions (Total Spent)	60

Top 3 Categories by Revenue

Rank	Category	Revenue
------	----------	---------

|-----|-----|-----|

| 1 | Butchers | \$208,118 |

| 2 | Electric household essentials | \$203,813.5 |

| 3 | Beverages | \$197,047.5 |

Top 3 Payment Methods by Revenue

| Rank | Payment Method | Revenue |

|-----|-----|-----|

| 1 | Cash | \$537,710 |

| 2 | Digital Wallet | \$507,279 |

| 3 | Credit Card | \$507,082 |

6. Key Insights & Strategic Recommendations

■ Business Insights

- **Category concentration:** Over a third of revenue stems from just three categories. Targeted promotions or inventory optimization in these areas could boost margins.
- **Payment mix:** Cash dominates but digital wallets are rapidly catching up. Enhancing digital wallet incentives may capture higher value, tech-savvy shoppers.
- **Discount impact:** One-third of purchases use a discount, yet the discount column is missing for the same proportion of rows. This suggests either data capture gaps or that discounts are often not recorded—standardizing discount logging is advisable.

■■ Data Quality Concerns

- **Missing critical fields** (*Item, Price Per Unit, Quantity, Total Spent*) affect ~5-10 % of rows. Imputation or source-system fixes are needed before building predictive models.
- **Outlier transactions** (60 IDs) could represent data entry errors, high-value bulk sales, or fraud. Conduct a manual review or apply a rule-based flagging system.

■ Recommendations

1. Data quality improvement plan

- Implement mandatory field validation at data entry (especially for *Item, Price Per Unit, Quantity, Total Spent*).
- Standardize the capture of discount information; ensure a boolean flag is always recorded.

2. Revenue growth tactics

- Run **category-specific loyalty offers** for the top-earning categories to increase repeat purchases.
- Introduce **cashless incentives** (e.g., 1 % rebate for digital-wallet payments) to shift a larger share of revenue to lower-cost payment channels.

3. Outlier investigation

- Cross-reference the 60 outlier transaction IDs with source logs (e.g., POS receipts) to confirm legitimacy.
- If many are erroneous, develop an automated cleansing rule (e.g., cap *Total Spent* at a reasonable percentile).

4. Analytics roadmap

- After cleaning missing values, build **customer-segmentation** models (RFM analysis) to personalize marketing.
- Leverage the clean **Transaction Date** field for **seasonality** and **trend** analysis (monthly/quarterly revenue patterns).

Prepared by the autonomous AI Data Analyst.